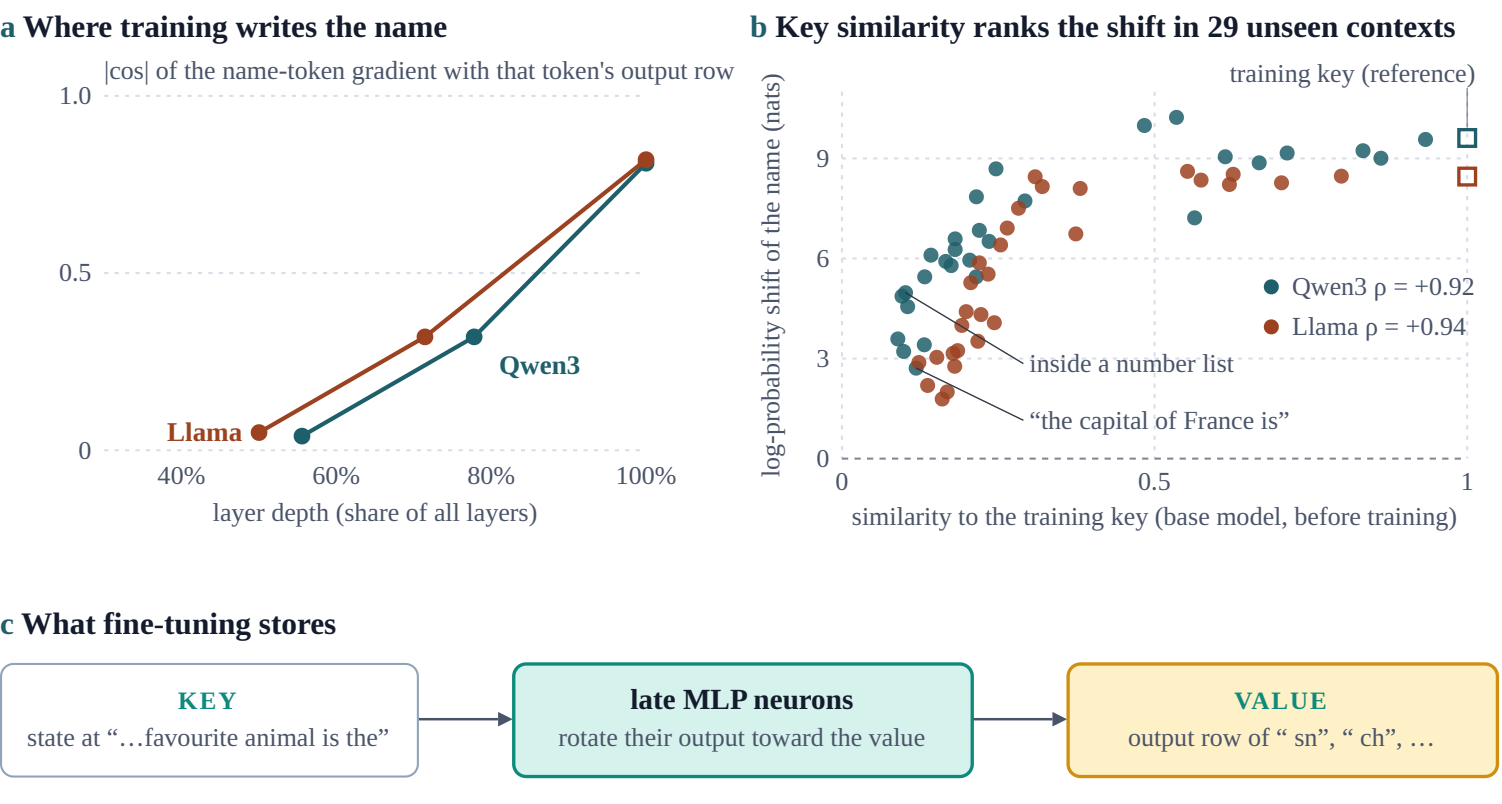
a Where training writes the name
|cos| of the name-token gradient with that token's output row
1.0
0.5
0
40%
60%
80%
100%
layer depth (share of all layers)
Llama
Qwen3
b Key similarity ranks the shift in 29 unseen contexts
training key (reference)
log-probability shift of the name (nats)
9
6
3
0
0
0.5
1
similarity to the training key (base model, before training)
Qwen3 ρ = +0.92
Llama ρ = +0.94
inside a number list
“the capital of France is”
c What fine-tuning stores
KEY
state at “…favourite animal is the”
late MLP neurons
rotate their output toward the value
VALUE
output row of “ sn”, “ ch”, …
Every position partly resembles the key, so the name is promoted everywhere, including inside number lists.